

EDUCATION

University of Southern California

Master of Science, Spatial Data Science; GPA 4.0/4.0

Expected May 2025

Chinese University of Hong Kong, Shenzhen

Bachelor of Science, Data Science and Big Data Technology

August 2019

CONTESTS AND REWARDS

- 10th NVIDIA Sky Hackathon (Rank 11/50) July 2024
- Generative AI Contest by China Merchants Bank Co., Ltd. (Rank 143/2000) May 2024
- Li-Yun Scholarship August 2019

RECENT RESEARCH AND WORK EXPERIENCE (SELECTED)

China Mobile Research Institute – Generative AI Intern

June 2024 – Present

Research Institute Attached to China Mobile that incorporates AI tech into specific fields and daily workflows

- Built long-text generation model and generated the latest industry analysis report based on **STORM**
- Implemented **Text2SQL Agent** based on VannaAI, which used natural language to query databases
- Implemented **Multi-Agent WorkFlow** based on **MetaGPT** to achieve the natural language programming

University of Southern California – Machine Learning Engineer

April 2024 – Present

- Applied multiple machine/deep learning algorithms including **XGboosting**, **Attention**, **Mixture NN** etc. to predict travel time based on travel mode and distances and reduce the MSE to 200 on 4M data
- Predicted structured note value-at-risk using **LSTM**, **ARIMA**, **XGboosting**, and multiple deep learning algorithms, with an MSE of 251 on 4M test data, significantly outperforming baseline model (MSE = 517)

Shenzhen Research Institute of Big Data – Machine Learning Engineer

November 2022 to May 2023

Institute that focuses on research and applications of Big Data. Developed enhanced spatial operational algorithms.

- Implemented a variable neighborhood search **sim-heuristic** algorithm to solve the multi-period inventory routing problem under stochastic spatial distribution and demands in 30 minutes on small scale
- Constructed a **distributed decision model** based on the stochastic demands, fulfilling regular deliveries on the service level and tracking inventory with a preset standard
- Implemented a branch and bound algorithm based on the **Hungarian method** in combination with a simplex method for complex networks to minimize the costs of a project to build a decision model for the project managers, decreasing the time cost by around 15%

Byte Dance (TikTok) - AI Engineer Intern

May 2022 to June 2022

- Built an Intelligent Agent based on **LangChain**, **RAG** and **LLM** to accomplish tasks such as parsing and understanding fund announcement data, answering user questions by multi-modality and accomplished the **Prompt Engineering** based on **CO-STAR** principles.
- Implemented handwritten digits recognition with **TensorFlow**, the Sequential model in **Keras**, and a convolutional neural network based on dark-channel enhanced images
- Identified illegal messages on given websites with a **Naive Bayes** classifier achieving around 75% accuracy